

Linear Algebra—Winter/Spring 2026

Instructors: A.S. Arora, S. Das, G. R. Kurri

Problem Set 8

(Updated: 12:50 PM, Wednesday, March 18, 2026)

Due: 10 PM, Tuesday March 24, 2026

Instructions

- a. *Timeline.* In general, expect the first “half” of a problem set to be released by Tuesday nights (or whenever the first lecture of the week takes place). Expect the full problem set to be released by Friday nights (or whenever the second lecture of the week takes place). The full assignment will, *typically*, be due on Tuesday nights, 8 PM.
- b. *Punctuality.* Aim to get the assignment done three days prior to the deadline. We will not be able to entertain any requests for extensions. If you seek clarification from us and we do not respond on time, do the following: (a) assume whatever seems most reasonable, and solve the question under that assumption and (b) ensure you have evidence for making the clarification request and report the incident to one of the PhD TAs.
- c. *Resources.* Feel free to use any resource you like (including generative AI tools) to get help but please make sure you understand what you finally write and submit. The TAs may ask you to explain the reasoning behind your response if something appears suspicious.
- d. *Graded vs Practice Questions.* Problems numbered using Arabic numerals (i.e. 1, 2, 3, ...) constitute assignment problems that will be graded. Problems numbered using Roman numerals (i.e. i, ii, iii, ...) are practice problems that will not be graded.

The text here is Linear Algebra Done Right, by Sheldon Jay Axler (3rd edition); we are on Chapter 5 (but basic material has been taken from Chapter 3).

We may add a few more practice questions later, but we do not plan to change the graded questions.

-
1. Suppose $U_1, \dots, U_m$ are finite-dimensional subspaces of a vector space V such that $U_1 + \dots + U_m$ is a direct sum. Prove that $U_1 + \dots + U_m$ is finite-dimensional and

$$\dim(U_1 \oplus \dots \oplus U_m) = \dim U_1 + \dots + \dim U_m.$$

2. Let $T \in \mathcal{L}(\mathbb{F}^3)$ be the linear operator defined by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3).$$

- a. Find all eigenvalues of T .
 - b. For each eigenvalue, determine the corresponding eigenspace.
3. Determine whether the following statement is true or false. Justify your answer.

Suppose V is a vector space. Let $A \subseteq V$ be a set of linearly independent vectors and $B \subseteq V$ be another set of linearly independent vectors such that $|A| < |B|$. Then there exists $x \in B \setminus A$ such that $A \cup \{x\}$ is also a linearly independent set.

4. Suppose $T \in \mathcal{L}(V)$ and $S \in \mathcal{L}(V)$ is invertible.
- a. Prove that T and $S^{-1}TS$ have the same eigenvalues.
 - b. What is the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$?
5. Provide a proof of the following statement without using any results about upper triangular matrices.

If a linear operator $T \in \mathcal{L}(V)$ has a diagonal matrix with respect to some basis of V , then the entries along the diagonal are precisely the eigenvalues of T .

6. Suppose $F = \mathbb{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbb{C})$ is a polynomial, and $\alpha \in \mathbb{C}$. Prove that α is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for some eigenvalue λ of T .
- i. Show that the result in Problem 6 does not hold if $\mathbb{C}$ is replaced with $\mathbb{R}$ everywhere.
 - ii. Prove or disprove the following.

If U_1, U_2, W are subspaces of a vector space V such that $V = U_1 \oplus W = U_2 \oplus W$, then $U_1 = U_2$.